

Ten Layers from Hilbert–Schmidt Mass to a Projection-Free Spectral Tail

Bin Wang

July 2026

Abstract

RH-282–RH-291 reopen the noisy determinant route by replacing a poorly-conditioned physical Riesz quotient with a normal diagonal realization of the exact projection-free spectral factor. A modulus-complete noisy head and the sharp small-noise Hilbert–Schmidt mass law instantiate the RH-279 variable-rank tail on the radius-7/5 target disk. The logarithmic block clock is sharp for the available mass information, the head is minimal once its cutoff is fixed, and the moving high-order tail factor of the complementary $\det_2$ product converges to one with every fixed derivative. Separately, the correct monodromy comparison uses the finite radius β_k ; this improves every archived cloud row but does not turn seven floating endpoints into an asymptotic theorem. A synchronized growing trace prefix exists, while a hidden-shell counterexample proves that unweighted prefix agreement cannot identify a divisor. The spectral and graded branches each satisfy four of five obligations, but the direct weighted complement-to-anchor bridge is absent. Gates A–E remain open.

1 The reopened tail

Let $(\mu_j(\sigma))$ be the algebraic bulk spectrum of $A_\sigma = K_\sigma/0.85$ after the two peripheral roots are removed. RH-276 gives

$$\sum_j |\mu_j(\sigma)|^2 \leq \|A_\sigma\|_2^2 \leq \sigma^{-1}$$

for sufficiently small noise. RH-282 selects every root with $|\mu_j| > 1/2$ and realizes the rest as a normal diagonal operator C_σ . At $R = 7/5$ and $m_\sigma = \lceil 4 \log(1/\sigma) \rceil$,

$$\limsup \|C_\sigma^{m_\sigma}\|_1^{1/m_\sigma} R \leq \frac{7}{10} e^{1/4} = 0.898817 \dots < 1.$$

This is the first archived instantiation of the RH-279 tail conditions.

RH-283 proves that the critical block slope for mass exponent α , cutoff q , and radius R is

$$a_{\text{crit}} = \frac{\alpha}{\log(1/(qR))}.$$

Equality gives logarithmic decay, and a diagonal saturation family proves failure below it. RH-284 makes the head canonical relative to q : it is the unique smallest spectral multiset leaving complement radius at most q . RH-285 upgrades the trace tail to every fixed derivative of the logarithmic canonical product.

2 The corrected bridge target

The exact RH-17 shell has Hardy-scaled radius

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right], \quad \beta = (0.85\sqrt{\lambda})^{-1}.$$

RH-286 proves

$$2(k-1)|\beta_k - \beta| \rightarrow |\beta| \log C_M, \quad 2|\beta_k^n - \beta^n| = O(k^{-1})$$

for fixed n . Root- ℓ^1 centering at the limiting radius therefore contains an accumulated radial term that fixed moments do not. Re-centering the seven RH-15 clouds at β_k reduces the total root-error range from 0.6424–1.2481 to 0.2841–0.8992. These values remain floating diagnostics.

RH-287 synchronizes fixed-order noisy trace convergence with the finite shell to obtain a diverging prefix. RH-288 shows that determinant gluing requires the stronger weighted prefix budget

$$P_\sigma(R) = \sum_{2 \leq n < m_\sigma} \frac{|\tau_{\sigma,n} - a_n| R^n}{n} \rightarrow 0,$$

where $\tau_{\sigma,n}$ is the modulus-complement trace. If $c_{\sigma,n} = h_{\sigma,n} + \tau_{\sigma,n}$ is the total noisy trace and $s_{k_\sigma,n}$ is the counterloop moment, then

$$\tau_{\sigma,n} - a_n = (c_{\sigma,n} - s_{k_\sigma,n} - a_n) - (h_{\sigma,n} - s_{k_\sigma,n}).$$

Thus a sufficient route must control both parenthesized errors with weights on the same logarithmic clock. RH-287 controls only the first one unweighted on an unspecified growing clock. RH-289 explains the gap: a complete root-of-unity shell can be invisible to any prescribed finite prefix while adding the exact divisor $1 - (\gamma z)^L$ just beyond it.

3 Typed frontier

branch	head	bridge	tail	target	boundary	score
noisy modulus spectrum	1	0	1	1	1	4
graded monodromy counterloop	1	1	0	1	1	4

The coordinatewise maximum is the all-ones vector, but RH-290 proves that this merge is ill typed until the direct weighted complement-to-anchor prefix, or both terms in its typed decomposition, are available. Both complete counts are zero.

Proposition 3.1 (next executable leaf). *The remaining determinant leaf is a direct weighted prefix theorem from the actual modulus-complement traces to the deterministic numerator anchor. A sufficient two-estimate realization consists of a weighted total-noisy-trace bridge to counterloop plus anchor and weighted modulus-head transport to the finite-radius counterloop, both on the RH-282 logarithmic clock. The spectral complement and target tails no longer need replacement.*

Proof. RH-285 supplies the noisy high-order tail, and the deterministic envelope batch supplies the target tail. RH-288 then reduces uniform determinant gluing exactly to the direct complement-to-anchor prefix budget and gives the displayed two-error decomposition. RH-286 identifies the correct finite-radius target for the counterloop term. $\square$

Remark 3.2. The normal spectral realization is an exact determinant-factor device, not a well-conditioned physical Riesz compression. No Gate A–E status changes. Nothing in this batch constructs a Hilbert–Polya operator, identifies Riemann zeros, proves a von Mangoldt trace formula, a completed-zeta divisor identity, or RH.